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Palaeointensity just at the onset of the Cretaceous normal superchron

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ABSTRACT

A combined palaeomagnetic and geochronologic study has been conducted on a basaltic lava sequence of five flows at Jianchang in Liaoning Province, northeastern China. Radiometric $^{40}Ar/^{39}Ar$ dating indicates that the volcanism occurred at about 119 Ma within the marine anomaly C34n in Cretaceous normal superchron (CNS). Rock magnetic investigations show that pseudo-single domain (PSD) or a mixture of single domain and multi-domain grain titanium-poor titanomagnetite is dominant in the studied lava flows. Both stepwise thermal and alternating field demagnetizations isolate the well-defined normal characteristic remanent magnetization (ChRM) in five independent lava flows with a mean direction of $D/I = 11.1°/57.2°$ ($\alpha_{95} = 5.7°$). Palaeointensity was determined using the Coe modified Thellier method with systematic partial thermoremanent magnetization (pTRM) checks. All together, 61 samples from 286 samples with positive pTRM and tail checks yield palaeointensities ranging between 10.1 and 73.6 μT. To further check the effects of thermal alteration on the palaeointensity results, a new ratio ($M_{rs,480°C}/M_{rs,25°C}$, where $M_{rs,25°C}$ and $M_{rs,480°C}$ are the room temperature saturation isothermal remanent magnetizations for the raw sample and the product after a 480 °C thermal treatment, respectively) is proposed. Results show that the ratio changes significantly even for those samples with positive pTRM check. We further put forward that palaeointensity results are acceptable with ratio between 0.9 and 1.1. With this new criteria, 17 samples show consistent palaeointensity results with an average of 20.0 ± 3.2 μT. This value corresponds to the virtual dipole magnetic moments (VDM) of (3.5 ± 0.6) × 10^{22} Am^2, which is about half of the value of present field. This finding, combined with other available results, suggests that magnetic field of the Earth just at the onset of the CNS is characterized by a weak palaeointensity.
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1. Introduction

The long-term variations in the strength of the geomagnetic field provide important constraints on the chemical–physical processes of the Earth's interior (Prévot and Perrin, 1992; Coe et al., 2000). Especially, the intensity of the geomagnetic field during the Cretaceous normal superchron (CNS) is crucial to understand the geodynamo. Glatzmaier et al. (1999) have suggested that the Earth's dipole moment correlates negatively with the geomagnetic reversal rate, which implies that the geomagnetic field during the CNS should be strong. However, there exist arguments about the measured palaeointensity data, e.g., high palaeointensities obtained from plagioclase single crystals (Tarduno et al., 2001, 2002) versus low palaeointensities from whole rocks and submarine basalt glass (SBG) (e.g., Prévot et al., 1990; Pick and Tauxe, 1993; Tanaka and Kono, 2002; Riisager et al., 2003; Zhu et al., 2004a,b, 2008; Pan et al., 2004; Zhao et al., 2004; Shi et al., 2005). Although it has been reported that palaeointensities near the onset and the termination of the CNS are only half of the present day field strength (Pick and Tauxe, 1993), debates still remain due to the insufficient palaeointensity data (Goguitchaichvili et al., 2002a; Ruiz et al., 2006). Therefore, the palaeointensity during the geomagnetic field switching from a reversing to a non-reversing state at the onset of the CNS is particularly interesting. In this paper, we present the first palaeointensity results obtained from Jianchang basalts, NE China, erupted at ~119 Ma.

2. Geological setting

Mesozoic volcanic rocks are extensively distributed in west Liaoning Province and its adjacent areas. Previous geochronologic and geochemical investigations in this region showed that the main eruptions occurred during four major eruption periods from the early Jurassic to early Cretaceous: (1) the Xinglonggou formation, 180–170 Ma; (2) the Lanqi formation, 160–150 Ma; (3) the Yixian formation, 126–120 Ma; and (4) the Yingchengzi formation, 110–100 Ma (Chen et al., 1992; Yang and Li, 2008). These volcanic activities also show an overall eastward temporal–spatial migration (Chen and Chen, 1997; Wang et al., 2006). The Early Cretaceous was the most strongest volcanism episode in west Liaoning, which consists mainly of weak alkalic medium–basic extrusives, breccias and tuffs. Olivine basalts and andesite-basalts dominate some intervals. They erupted in a tectonic extensional setting, as

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indicated by the contemporary occurrence of A-type granite, dolerite dyke swarms and metamorphic core complexes. It is proposed that this giant igneous event is related to the coeval lithospheric delamination in eastern China, which may be resulted from the Kula-Pacific Plate subduction, possibly aided by major superplume activity associated with global-scale mantle upwelling (Wu et al., 2005). Previous palaeomagnetic investigations at the Fuxin-Yixian basin showed that west Liaoning had experienced regional deformation without significant change in latitude displacement since the Early Cretaceous (Zhu et al., 2001, 2004a).

The studied section (40°50.32′N, 119°46.53′E) is about 2 km west of the Jianchang county (Fig. 1). Five lava flows with a total thickness of about 35 m are exposed along a road-cut section. No sedimentary layers are observed between them. Horizons with some bubbles or weathered horizons in violet colors were used to distinguish different flows in the field. At least 30 fresh cores were taken from each flow with a portable gasoline-powered diamond drill from each lava flow (270 orientated cores in total) from two fieldworks. Cores were orientated with magnetic and sun compasses (ASC Scientific, California). The samples, which were marked with capital letter T (JC) for initials, were collected at the first (second) fieldwork. Four sites from upper to lower part of the flow were distributed in Lava 1 (from site 1 to 4) and Lava 3 (from site 8 to 11) with interval of about 5–8 m. Three sites were in Lava 2 (from site 5 to 7) and two sites were in Lava 4 (from site 12 to 13). Only one site was collected from Lava 5 (site 14). So far no radiometric dating has been reported on these volcanic rocks. Fresh massive samples, which were marked with sequence of site, were collected from each lava flow for radiometric dating.

3. $^{40}Ar/^{39}Ar$ dating

Basalt samples were crushed and sieved between 40 and 80 mesh (380–200 μm) and were washed with distilled water. Volcanic matrixes without phenocrysts were hand-picked under binocular microscope and were washed with acetone in an ultra-sonic bath for 20 min. To eliminate possible alterations, matrix samples were washed with 5% HNO_3 in an ultra-sonic bath for 20 min. Then the grains were rinsed with distilled water, dried, wrapped in aluminum foil and irradiated together with TCR-sanidine standards, optical CaF_2 and K-glass monitors in position H8 of the 49-2 reactor, Beijing, China, for 23.5 h with a 0.5 mm cadmium foil shield. The reference age for TCR-sanidine is 28.34 ± 0.28 Ma (Renne et al., 1998). Ca, K correction factors were calculated from the CaF_2 and K-glass monitors: $(^{40}Ar/^{39}Ar)_K = 8.8 \times 10^{-4}$, $(^{39}Ar/^{37}Ar)_{Ca} = 7.92 \times 10^{-4}$, $(^{36}Ar/^{37}Ar)_{Ca} = 2.21 \times 10^{-4}$. The total fusion of standards and step-heating analysis of samples were performed at the $^{40}Ar/^{39}Ar$ Geochronology Lab, Institute of Geology and Geophysics, Chinese Academy of Sciences, on a MM5400 mass spectrometer operating in static mode. Data were corrected for system blanks and mass discriminations suggest argon isotopes derived from Ca, K and decay of ^{37}Ar since the time of irradiation. The decay constant (λ) used throughout the calculations is $(5.543 \pm 0.010) \times 10^{-10}$ a^{-1}, as recommended by Steiger and Jäger (1977). The detailed analytical and data processing procedures refer to He et al. (2004). The results of the $^{40}Ar/^{39}Ar$ step-heating are shown in Table 1 and are plotted as age spectrum and the inverse isochron in Fig. 2.

The matrix of T5 yield a concordant age spectrum (Fig. 2a). Nine consequent steps, which account for 82.2% of the total ^{39}Ar released, define a plateau age of 119.3 ± 1.2 Ma (2σ). Inverse isochronal age 118.9 ± 1.4 Ma (2σ, MSWD = 0.4), calculated from nine steps that formed the plateau (Fig. 2b), is in good agreement with the plateau age. The $^{40}Ar/^{36}Ar$ intercept of 315.8 ± 31.8 (2σ) is not distinguishable from the air ratio, indicating that there's no apparent excess argon contamination (Fig. 2b). Alteration index of three plateau steps are below cut off value, 0.0006, indicating little alteration (Baksi, 2007). Therefore, 118.9 ± 1.4 Ma (2σ) is the good estimate of the eruption age of lava T5.

The matrix from sample T8 gives a slightly disturbed age spectrum (Fig. 2c). Despite this, the eight consequent steps, which

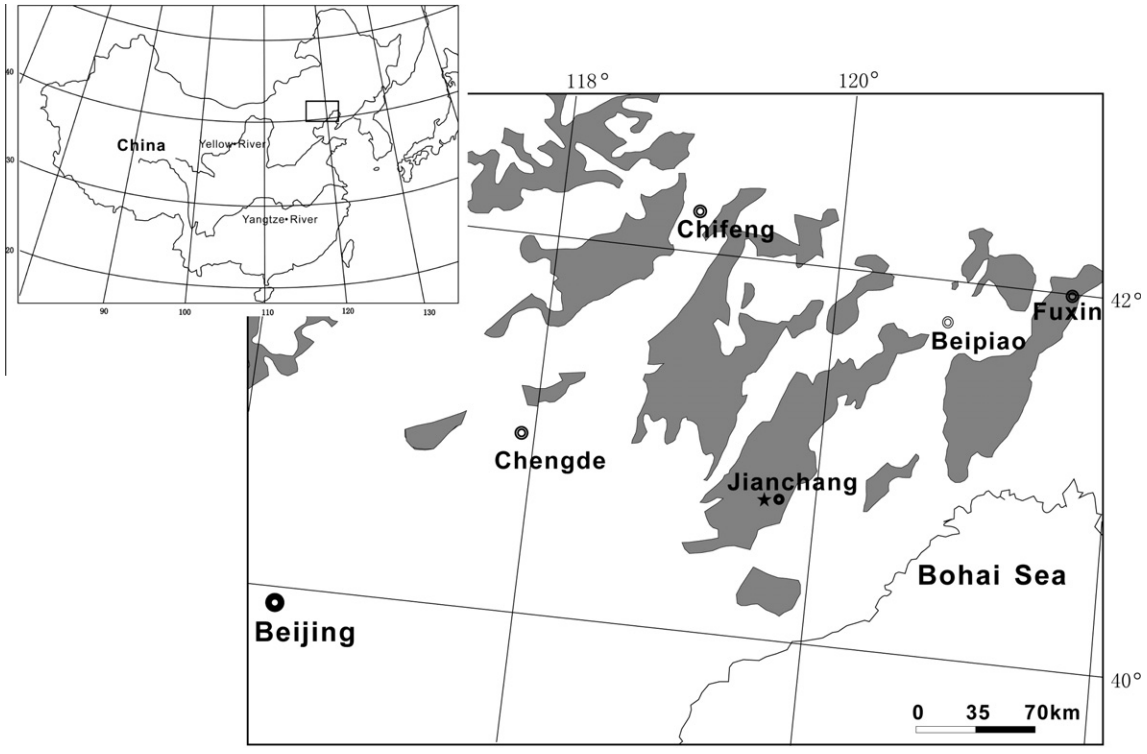


Fig. 1. A schematic map showing the Cretaceous rock distribution in the studied area. The star symbol indicates the sampling site at Jianchang.

Table 1
$^{40}Ar/^{39}Ar$ analytical data for the selected samples.

Temp. (°C)	$^{39}Ar/^{40}Ar \pm 1$ s.d. (10^{-2})	$^{36}Ar/^{40}Ar \pm 1$ s.d. (10^{-4})	$^{37}Ar/^{39}Ar$	^{39}Ar Cum (%)	$^{40}Ar^{*}$ (%)	$^{40}Ar^{*}/^{39}Ar$	A.I.	Apparent age ± 2 s.d. (Ma)
Matrix separate from T5, weight = 3.79 mg, J = 0.005096 ± 0.000026								
720	4.76 ± 0.09	9.17 ± 0.47	0.34	1.06	72.89	15.32	0.00982	135.91 ± 3.79
770	6.52 ± 0.10	2.45 ± 0.14	0.32	2.27	92.75	14.22	0.00191	126.50 ± 1.95
820	6.37 ± 0.07	4.10 ± 0.09	0.33	5.07	87.88	13.80	0.00328	122.88 ± 1.51
850	6.39 ± 0.08	4.65 ± 0.12	0.30	7.33	86.26	13.51	0.00371	120.37 ± 1.68
880	7.25 ± 0.10	0.85 ± 0.05	0.23	10.05	97.48	13.44	0.00060	119.81 ± 1.58
900	7.33 ± 0.07	0.61 ± 0.05	0.21	9.81	98.19	13.39	0.00042	119.38 ± 1.17
920	7.32 ± 0.08	0.74 ± 0.04	0.23	9.48	97.81	13.36	0.00051	119.14 ± 1.23
950	7.30 ± 0.08	0.89 ± 0.07	0.26	10.28	97.36	13.34	0.00062	118.90 ± 1.29
980	7.24 ± 0.07	1.17 ± 0.07	0.30	8.41	96.54	13.34	0.00082	118.95 ± 1.18
1030	7.16 ± 0.07	1.50 ± 0.05	0.36	10.52	95.55	13.34	0.00107	118.91 ± 1.19
1080	7.03 ± 0.07	2.04 ± 0.06	0.39	10.61	93.96	13.36	0.00148	119.08 ± 1.24
1150	6.97 ± 0.08	2.08 ± 0.11	0.46	5.71	93.84	13.46	0.00152	120.01 ± 1.48
1250	5.86 ± 0.09	6.47 ± 0.31	0.61	2.19	80.88	13.80	0.00562	122.85 ± 2.47
1400	6.19 ± 0.07	5.66 ± 0.18	0.37	7.21	83.27	13.44	0.00466	119.83 ± 1.54
Matrix separate from T8, weight = 4.01 mg, J = 0.005087 ± 0.000025								
750	5.06 ± 0.03	8.53 ± 0.27	1.45	3.12	74.79	14.78	0.00857	131.08 ± 1.55
800	6.76 ± 0.06	2.26 ± 0.08	2.23	4.49	93.32	13.80	0.00170	122.65 ± 1.02
840	5.61 ± 0.04	7.88 ± 0.17	2.08	4.24	76.71	13.68	0.00715	121.63 ± 1.15
880	7.16 ± 0.04	1.12 ± 0.10	0.43	8.03	96.69	13.50	0.00079	120.10 ± 0.76
920	7.29 ± 0.04	0.74 ± 0.05	0.29	10.04	97.80	13.42	0.00052	119.42 ± 0.59
1000	7.19 ± 0.04	1.25 ± 0.06	0.25	10.54	96.31	13.40	0.00088	119.27 ± 0.64
960	7.28 ± 0.05	0.82 ± 0.05	0.28	10.69	97.57	13.40	0.00057	119.25 ± 0.83
1040	7.11 ± 0.04	1.35 ± 0.06	0.21	8.55	96.00	13.51	0.00097	120.18 ± 0.66
1080	7.04 ± 0.03	1.98 ± 0.08	0.25	11.41	94.14	13.38	0.00143	119.09 ± 0.61
1120	6.99 ± 0.04	2.01 ± 0.07	0.34	9.75	94.05	13.45	0.00146	119.68 ± 0.77
1200	7.18 ± 0.04	1.37 ± 0.05	0.39	12.39	95.96	13.37	0.00097	119.03 ± 0.62
1300	7.07 ± 0.08	1.32 ± 0.14	0.38	4.46	96.09	13.59	0.00095	120.91 ± 1.49
1400	7.03 ± 0.09	1.62 ± 0.16	0.66	2.29	95.19	13.55	0.00118	120.53 ± 1.62
Matrix separate from T10, weight = 3.64 mg, J = 0.005134 ± 0.000026								
740	5.35 ± 0.08	6.23 ± 0.29	0.45	1.37	81.58	15.25	0.00598	136.32 ± 2.35
780	6.87 ± 0.09	1.49 ± 0.14	0.39	2.46	95.58	13.91	0.00111	124.72 ± 1.65
820	4.65 ± 0.02	12.41 ± 0.25	0.43	5.13	63.32	13.62	0.01370	122.19 ± 1.51
860	6.27 ± 0.03	5.53 ± 0.13	0.38	8.32	83.66	13.33	0.00452	119.74 ± 0.75
890	7.33 ± 0.03	0.91 ± 0.06	0.33	11.52	97.32	13.28	0.00063	119.23 ± 0.56
920	7.41 ± 0.03	0.55 ± 0.04	0.28	11.48	98.35	13.27	0.00038	119.19 ± 0.56
980	6.92 ± 0.03	2.84 ± 0.06	0.23	11.58	91.61	13.24	0.00210	118.92 ± 0.55
1020	7.27 ± 0.03	1.38 ± 0.04	0.20	12.35	95.91	13.20	0.00097	118.57 ± 0.57
1060	7.13 ± 0.04	1.94 ± 0.06	0.17	9.74	94.25	13.21	0.00140	118.68 ± 0.65
1120	7.10 ± 0.04	1.97 ± 0.06	0.25	9.06	94.16	13.26	0.00143	119.12 ± 0.70
1200	7.37 ± 0.03	0.91 ± 0.04	0.55	8.16	97.31	13.20	0.00063	118.58 ± 0.56
1300	6.88 ± 0.03	2.93 ± 0.10	1.04	5.07	91.33	13.27	0.00219	119.17 ± 0.68
1400	6.84 ± 0.05	3.04 ± 0.16	1.02	3.74	91.02	13.30	0.00228	119.44 ± 1.01

Note: All isotopes are corrected to Ca and K correction factors and decay time between the irradiation and measurements.

account for 81.4% of the total ^{39}Ar released, define a plateau age of 119.5 ± 1.2 Ma (2σ). An inverse isochronal age of 119.6 ± 1.6 Ma (2σ, MSWD = 2.0), calculated from eight steps that formed the plateau, is consist with the plateau age. The $^{40}Ar/^{36}Ar$ intercept of 288.9 ± 59.5 (2σ) is in agreement with the air ratio (Fig. 2d). Alteration index of two plateau steps are below cut off value (0.0006), indicating little alteration (Baksi, 2007), and 119.6 ± 1.6 Ma (2σ) to be a reliable estimate of the eruption age of lava T8.

The matrix of T10 yields a slightly disturbed age spectrum (Fig. 2e). Nine consequent steps, which account for 82.7% of the total ^{39}Ar released, define a plateau age of 119.0 ± 1.2 Ma (2σ). An inverse isochronal age of 118.9 ± 1.2 Ma (2σ, MSWD = 1.1), calculated from nine steps that formed the plateau, is consist with the plateau age (Fig. 2f). The $^{40}Ar/^{36}Ar$ intercept of 299.3 ± 18.7 (2σ) is in agreement with the air ratio. Alteration index of one plateau step are below cut off value (0.0006), indicating little alteration (Baksi, 2007), and 118.9 ± 1.2 Ma (2σ) is the good estimate of the eruption age of lava T10.

4. Rock magnetic experiments

Magnetic hysteresis loops and the stepwise backfield demagnetization of the saturation isothermal remanence (SIRM) were measured on the MMVFTB with a maximum field of 1.0 T. At least nine samples from each flow (a total of 81 samples) were carried out in experiments. The hysteresis parameters (saturation magnetization, M_s, saturation remanence, M_{rs}, coercivity, H_c) were obtained after correcting the high-field slopes. The remanent coercivity (H_{cr}) was obtained using the backfield demagnetization of M_{rs}. The Day-plot (M_{rs}/M_s versus H_{cr}/H_c) was further constructed to characterize the domain states of magnetic minerals in samples. Their ratios fall in the pseudo-single-domain (PSD) grain-size region (Day et al., 1977) (Fig. 3). The first-order-reversal-curves (FORC) diagrams were measured on four samples from each flow with Micromag 2900 AGM and Micromag 3900 VSM to further determine the domain states of the magnetic particles. The sample JC1-011 and T14-6 display obviously closed and elongated contours probably resulted from the single-domain (SD) grains (Fig. 4a and i). The contours of the sample T12-3 are considerably vertically spreaded indicating multidomain (MD) grains (Fig. 4g) (Roberts et al., 2000). And a lot of samples display large vertical spread but closed contour, it is possibly caused by PSD or SD mixture with MD grains.

Temperature dependence of magnetization, $J(T)$, was measured to determine the Curie temperature and possible magnetochemical changes at elevated temperatures. The $J(T)$ measurements were measured by heating small amount of powder (~300 mg) up to

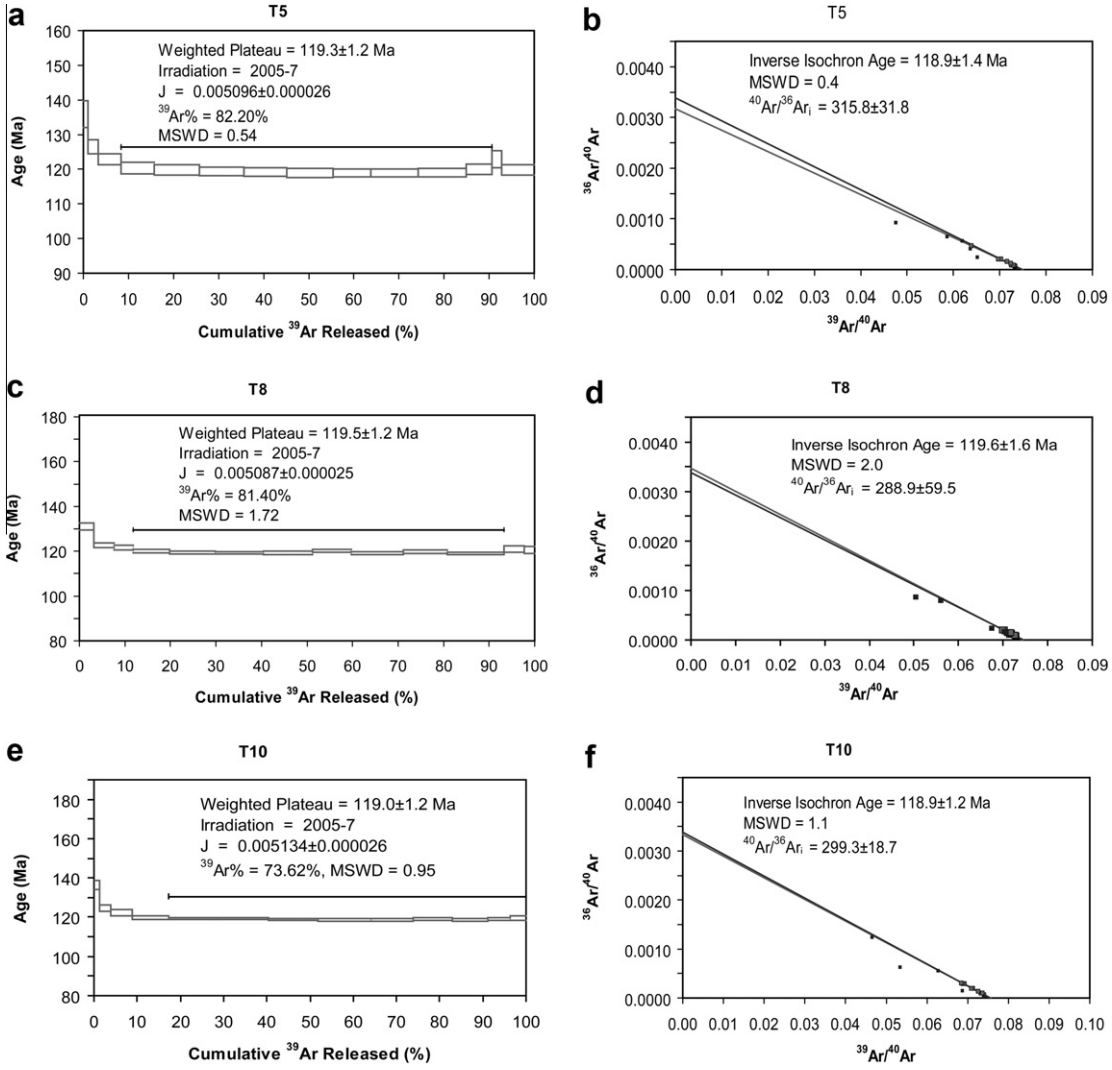


Fig. 2. (a) Age spectrum of matrix from sample T5. (b) Inverse isochronal plot of matrix from sample T5. (c) Age spectrum of matrix from sample T8. (d) Inverse isochronal plot of matrix from sample T8. (e) Age spectrum of matrix from sample T10. (f) Inverse isochronal plot of matrix from sample T10.

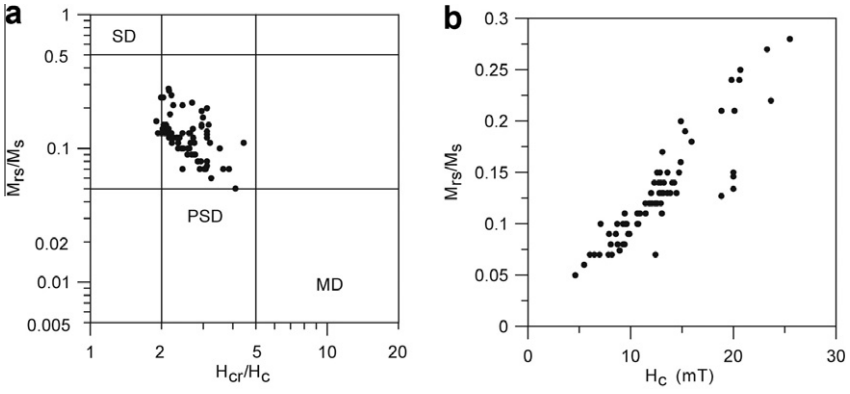


Fig. 3. Plot of hysteresis parameters of the studied samples, according to (a) Day et al. (1977) and (b) Tauxe et al. (2002). SD: Single-domain; PSD: pseudo-single-domain; MD: multidomain.

700 °C in an argon environment using MMVFTB equipped with a furnace. Majority samples exhibited nearly reversible $J(T)$ curves with Curie temperature of 550–585 °C (Fig. 5a–f), which indicates that the major magnetic mineral is Ti-poor titanomagnetite or stoichiometric magnetite.

The low temperature dependence of remanence was measured with MPMS-XL SQUID magnetometer. Samples were given a SIRM in a field of 2.5 T at room temperature, then in a zero field cooled to 5 K and warmed back to 300 K. The SIRM cooling and warming curves are shown in Fig. 5g and h. The remanence of sample T13-1 drops by 45% after the low temperature cycle, which is a characteristic feature of the PSD particles. The SIRM curve for T14-6 is almost flat and reversal during the cooling and warming cycle (Fig. 5h), which suggests the SD particles (Özdemir et al., 2002).

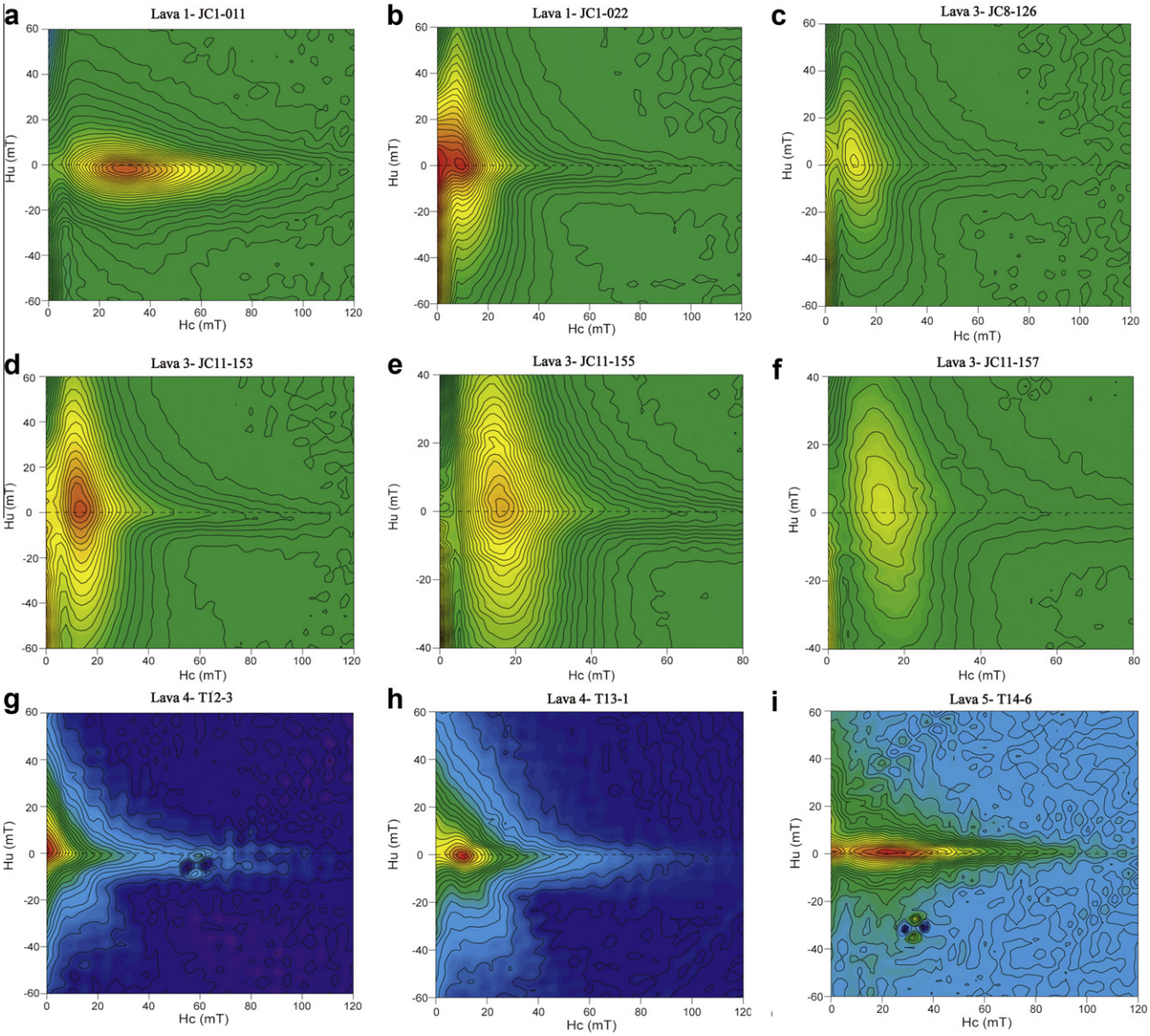


Fig. 4. FORC diagrams for representative samples. Smoothing factor = 3.

5. Palaeomagnetic direction

The cylinder samples with diameter of 2.5 cm were cut to shorter cylinders with length of 1.0 cm in the laboratory, and then the shorter specimens were further drilled to two parts: the small inner cylinders with diameter of 1.0 cm for palaeointensity experiments and the hollow rings for stepwise thermal or alternating field (AF) demagnetizations, respectively. At least 10 samples from each lava flow were subjected to thermal or AF demagnetizations. In most cases, stepwise thermal demagnetization up to 585 °C (Fig. 6) with temperature intervals of 25–50 °C was performed for palaeodirections by using an ASC TD-48 thermal demagnetizer with a residual magnetic field less than 10 nT as well as AF demagnetizations up to 60–80 mT (Fig. 5) with field intervals of 5–10 mT. Just five samples from Lava 5 were heated to 640 °C as maximum temperature. The remanence measurements were made in a field-free environment (<300 nT) by using 2G-755 SQUID cryogenic magnetometer with a degaussing system. Samples (191) from five lava flows were subjected to thermal demagnetization. Generally, soft viscous components were removed beyond 200 °C and only in a few cases a harder components persisted up to 300–350 °C. Above 350 °C univectorial components decreased linearly towards the origins. Six samples have been stepwise demagnetized by alternating field. As shown in Fig. 6, a soft component, which corresponds more or less with the low-temperature component, is removed after 10 mT, four out of six samples hold stable characteristic components isolated after 20 mT.

The direction of the stable characteristic remanent magnetization (ChRM) for each sample was determined by the principal component analysis (Kirschvink, 1980) and then was averaged for each lava flow. The results were summarized in Table 2. The mean directions of adjacent individual flows passed a *F*-test at 95% confidence (McFadden and Lowes, 1981) indicating that the eruptions of these five lavas were time independent. The overall averaged direction of five lava flows is $D/I = 11.1°/57.2°$ ($\alpha_{95} = 5.7°$) and the corresponding palaeomagnetic pole at 81.1°N and 224.2°E ($A_{95} = 7.6°$). The secondary low temperature components, removed by 200 °C (or highest temperature for 350 °C) have the averaged direction of $D/I = 359.8°/60.3°$, which is consistent with the present field direction of the sampling site, but with large

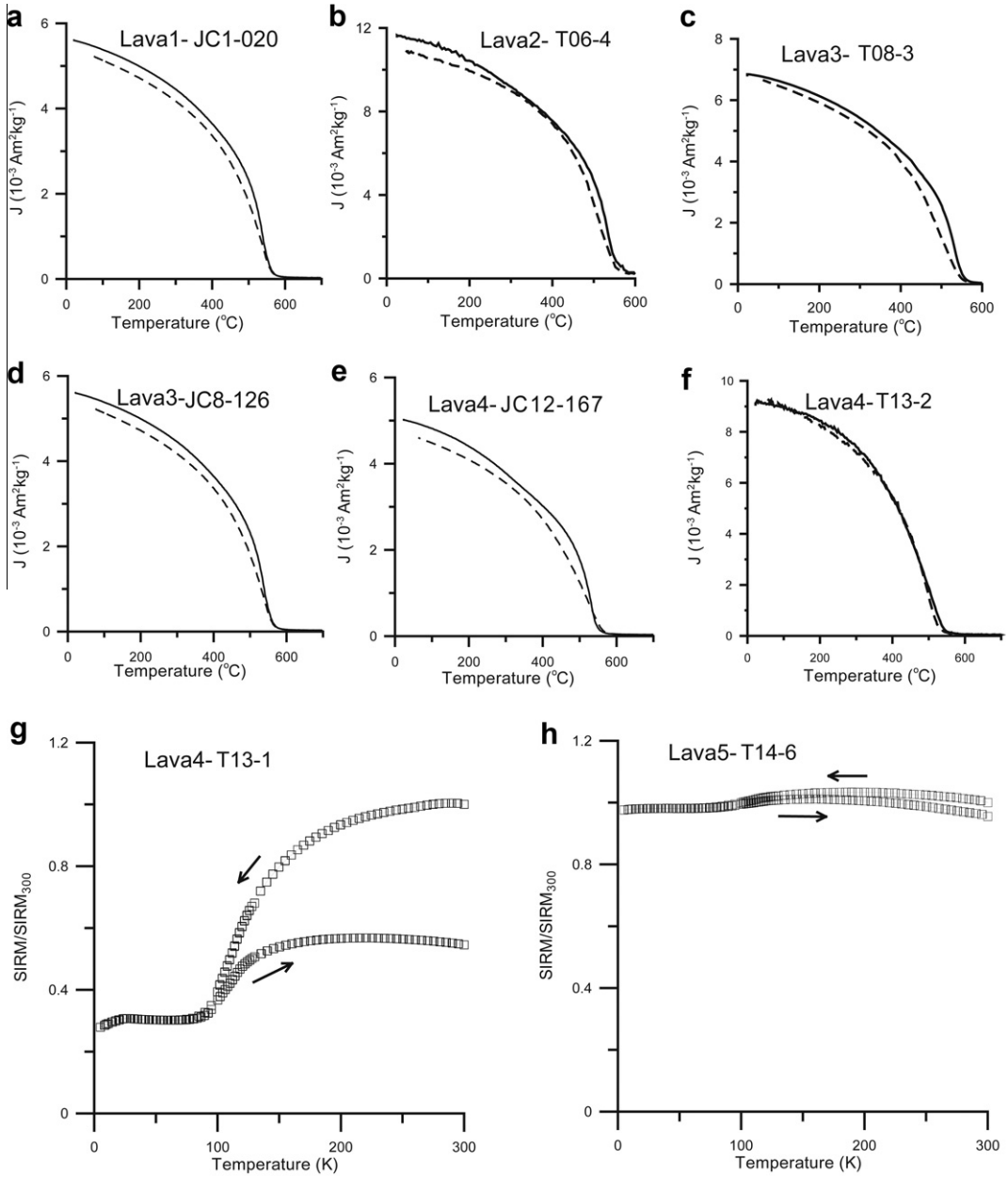


Fig. 5. (a–f) Representative results of thermomagnetic curves (*J*–*T*) of samples from each lava flow at Jianchang section. Solid and dashed lines stand for the heating and cooling curves, respectively; (g and h), normalized temperature variation of saturation isothermal remanence (SIRM) during zero field cooling from 300 to 5 K and zero field warming back to 300 K. SIRM was produced in a field of 2.5 T at 300 K.

scatter, thus are interpreted to be the viscous remanent magnetizations (VRM).

6. Palaeointensity determination

All together, 286 samples including some parallel samples were experimented for palaeointensity determinations. The Coe version (Coe, 1967) of the Thellier method with sliding pTRM checks (Prévot et al., 1985) and pTRM-tail checks (Riisager and Riisager, 2001) was used in this study. In order to perform the pTRM-tail check, samples were heated in zero field again after in-field heatings to the same temperatures. The standard pTRM checks were obtained after the pTRM-tail checks by heating to the desired previous lower temperatures and cooling in an applied field. The whole process was carried out in a shielded room (<300 nT). The temperature reproducibility between heatings to the same temperature was generally within 2 °C. Most importantly, during our experiments, argon gas was circulated slowly through the furnace during each heating and cooling cycle and charcoal powder was scattered on sample holder heating together with the samples to prevent possible oxidation during heating. The remanences were measured by using 2G760 SQUID magnetometer. The laboratory field for in-field heating and cooling was set to 30 μT (or 40 μT) with a precision better than 0.1 μT.

The Arai plots (Nagata et al., 1963) are divided into three categories. The first category shows linear behavior in certain temperature intervals. The second category shows concave-up NRM–TRM curves in Arai plots with positive pTRM check (<20%, and mostly

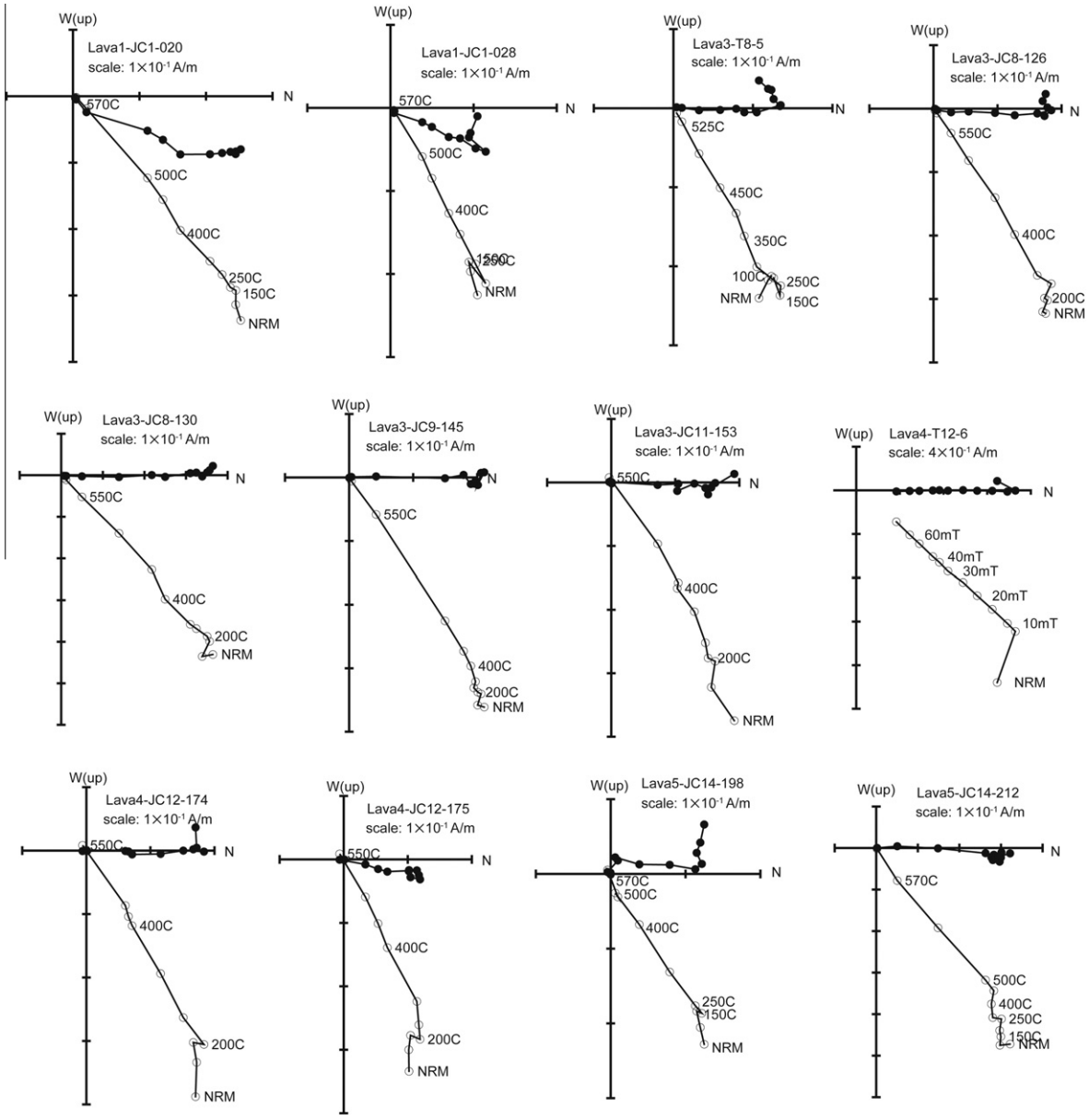


Fig. 6. Orthogonal projections of representative progressive thermal or AF demagnetization. The solid (open) symbols refer to the horizontal (vertical) plane.

Table 2
Palaeodirectional results from the Jianchang section.

Lava #	n/N	D (°)	I (°)	K	α_{95}	PLA (°)	PLO (°)	dp	dm
Lava 1	54/59	23.2	52.9	346.4	1.0	70.2	223.3	1.0	1.4
Lava 2	40/44	17.2	54.9	297.6	1.3	75.5	224.9	1.3	1.8
Lava 3	38/43	4.5	58.8	409.3	1.1	86.4	226.3	1.2	1.6
Lava 4	33/34	3.9	61.4	408.1	1.2	86.5	175.9	1.4	1.9
Lava 5	10/17	4.2	56.5	246.0	3.1	85.1	256.5	3.2	4.5
Mean		11.1	57.2	183.0	5.7	81.1	224.2	A_{95} = 7.6	

Note: n/N: The number of samples used in the palaeodirection calculation/total number of samples demagnetized; D/I: declination/inclination; α_{95}: radius of circle of the 95% confidence about the direction; PLA/PLO: VGP latitude/longitude; dp/dm: semi-axis of the confidence ellipse along/perpendicular to the site-to-pole great circle; A_{95}: radius of circle of the 95% confidence about the pole.

less than 15%). The last category is characterized by non-ideal Arai plot and negative pTRM checks probably resulted from the mineral alterations during the heating processes, and thus these results were excluded.

The first category samples were selected from at least four points in the NRM–TRM diagrams with linear behavior and with NRM fraction f larger than 0.3 to calculate the palaeointensity. The results were accepted with the positive pTRM checks (<15%) and pTRM tail checks (<10%), and with qualify factor q (Coe et al., 1978) mostly larger than 2.5 and the direction of demagnetization towards the origin. With these selection criteria, 61 out of 286 samples from four lavas yield acceptable palaeointensity results (Table 3). The Arai plots for representative samples were shown in Fig. 7. The palaeointensity results range from ~10.09 ± 0.48 to ~73.60 ± 6.75 μT.

7. Rock magnetism during palaeointensity experiments

The palaeointensity results vary significantly with a factor of ~4 even in the same lava flow. The concave-up NRM–TRM curves in Arai plots can be caused either by MD effects (Levi, 1977; Fabian, 2001; Shcherbakov and Shcherbakova, 2001; Biggin and

Table 3
Palaeointensity results from the Jianchang section.

Samples	n	T_{int}	f	g	q	$F \pm \sigma$ (μT)	$M_{rs,480°C}/M_{rs,25°C}$	$H_{c,480°C}/H_{c,25°C}$	Accept?	Fa ± σ_F (μT)	VDM ± σ_v (10^{22} Am^2)
JC1-005	4	510–590	0.37	0.63	9.4	38.43 ± 0.95	0.88	0.84	N	21.5 ± 0.9	4.0 ± 0.2
JC1-007	4	510–590	0.45	0.62	2.4	36.97 ± 4.34	0.83	0.81	N		
JC1-008	4	510–590	0.58	0.54	3.2	38.49 ± 3.83	0.88	0.90	N		
JC1-009	4	510–590	0.51	0.62	5.6	38.51 ± 2.16	0.88	0.72	N		
JC1-010	4	510–590	0.53	0.62	5.1	40.47 ± 2.64	0.81	0.78	N		
JC1-011	4	510–590	0.5	0.6	5.3	41.73 ± 2.31	0.75	0.68	N		
JC1-012	4	510–590	0.53	0.64	6.3	42.10 ± 2.27	0.67	1.11	N		
JC1-013	4	510–590	0.59	0.61	5.7	62.43 ± 3.90	0.67	0.82	N		
JC1-014	4	510–590	0.54	0.65	5.8	51.62 ± 3.09	0.63	1.01	N		
JC1-019	5	350–510	0.65	0.67	2.8	21.47 ± 3.33	1.34	1.62	N		
JC1-020	5	350–510	0.59	0.6	2.9	23.08 ± 2.78	0.98	1.07	Y		
JC1-021	5	350–510	0.45	0.61	2.4	19.48 ± 2.24	0.90	1.05	Y		
JC1-022	5	350–510	0.62	0.63	4.0	19.04 ± 1.82	0.97	1.03	Y		
JC1-023	5	350–510	0.49	0.62	4.0	22.95 ± 1.74	0.93	1.1	Y		
JC1-024	4	400–510	0.45	0.48	3.2	22.34 ± 1.49	1.01	1.13	Y		
JC1-025	5	350–510	0.42	0.56	3.2	20.40 ± 1.52	1.13	1.32	N		
JC1-028	5	350–510	0.39	0.69	3.0	20.01 ± 1.78	1.12	1.15	N		
JC1-029	5	350–510	0.34	0.68	2.7	23.51 ± 2.00	1.27	1.34	N		
T08-2A	4	400–510	0.5	0.66	3.6	30.28 ± 2.80	0.63	0.99	N	22.1 ± 2.4	3.8 ± 0.4
T08-3	6	300–520	0.31	0.75	3	21.34 ± 1.66	0.97	0.86	Y		
T08-5A	5	350–510	0.55	0.7	3.9	28.96 ± 2.87	1.02	1.01	Y		
JC8-126	5	350–510	0.41	0.73	3.8	69.97 ± 5.59	0.67	0.80	N		
JC8-127	6	350–530	0.49	0.79	3.2	57.11 ± 7.00	0.61	0.75	N		
JC8-128	5	350–510	0.44	0.71	3.4	73.60 ± 6.75	0.60	0.73	N		
JC8-129	5	350–510	0.45	0.73	4.2	62.85 ± 4.96	0.59	0.75	N		
JC8-130	5	350–510	0.42	0.73	4.3	65.52 ± 4.74	0.62	0.79	N		
JC9-145	5	350–510	0.4	0.66	2.8	16.89 ± 1.59	0.98	0.96	Y		
JC10-150	4	350–480	0.35	0.61	3.1	27.47 ± 1.89	1.08	1.08	Y		
JC10-151	5	350–510	0.61	0.53	6.3	14.60 ± 0.75	0.94	0.84	Y		
JC10-152	6	300–510	0.52	0.58	3.3	18.00 ± 1.63	0.79	0.82	N		
JC11-153	4	400–510	0.41	0.46	4.1	25.47 ± 1.19	1.02	0.97	Y		
JC11-155	5	350–510	0.41	0.69	6.9	18.77 ± 0.77	0.67	0.68	N		
JC11-156	5	350–510	0.54	0.69	4.5	18.17 ± 1.51	0.73	0.72	N		
JC11-157	5	350–510	0.53	0.67	3.5	50.32 ± 4.99	0.75	0.79	N		
JC11-158	6	350–530	0.59	0.73	4.6	39.47 ± 3.70	0.86	0.84	N		
JC11-160	7	300–530	0.5	0.76	4.5	46.47 ± 3.94	0.80	0.74	N		
JC11-161	6	350–530	0.69	0.67	6	15.29 ± 1.17	0.71	0.66	N		
JC11-162	5	350–510	0.52	0.72	4.5	28.48 ± 2.38	0.69	0.64	N		
JC11-162.5	7	350–560	0.73	0.52	5.6	16.11 ± 1.11	0.74	0.68	N		
JC11-163	5	350–510	0.44	0.55	3.1	18.34 ± 1.43	0.73	0.67	N		
T11-4	5	350–510	0.42	0.71	2.9	25.56 ± 2.64	0.76	0.71	N		
JC12-164	5	350–510	0.62	0.71	3.5	19.36 ± 2.43	0.74	0.68	N	16.4 ± 3.0	2.7 ± 0.5
JC12-167	5	350–510	0.62	0.52	10.1	14.76 ± 0.47	0.89	0.87	N		
JC12-168	5	350–510	0.64	0.61	4	14.71 ± 1.43	0.75	0.68	N		
JC12-174	5	350–510	0.59	0.42	16.1	15.17 ± 0.23	0.74	0.72	N		
JC12-175	5	350–510	0.59	0.59	4.1	15.85 ± 1.33	0.78	0.73	N		
JC12-177	5	350–510	0.53	0.65	2.9	16.57 ± 1.96	0.94	0.87	Y		
JC12-178	5	350–510	0.41	0.52	4.5	10.09 ± 0.48	1.05	0.96	Y		
T12-4	7	350–540	0.58	0.83	9.5	13.12 ± 0.67	1.09	1.04	Y		
T13-1A	5	350–510	0.47	0.74	2.6	27.39 ± 3.64	0.93	0.81	Y		
T13-1B	9	350–560	0.82	0.85	9.7	19.89 ± 1.42	1.01	0.96	Y		
T13-5B	5	400–540	0.33	0.71	1.4	34.27 ± 5.84	0.89	0.81	N		
T13-7	6	400–540	0.49	0.78	7.1	21.45 ± 1.16	1.34	1.24	N		
JC14-206	5	300–480	0.53	0.58	3.2	28.27 ± 2.77	0.58	0.58	N		
JC14-207	5	300–480	0.51	0.61	3.9	26.71 ± 2.11	0.63	0.61	N		
JC14-208	9	300–480	0.86	0.78	8.6	19.06 ± 1.50	0.66	0.60	N		
JC14-209	9	300–590	0.87	0.81	6.7	16.37 ± 1.75	0.67	0.63	N		
JC14-210	8	350–590	0.85	0.81	6.2	17.75 ± 1.96	0.58	0.55	N		
JC14-211	7	400–590	0.84	0.75	9.8	16.54 ± 1.06	0.65	0.63	N		
JC14-212	7	400–590	0.71	0.69	6.7	17.38 ± 1.26	0.73	0.67	N		
T14-6	6	350–540	0.24	0.77	3.8	32.10 ± 1.56	0.84	0.79	N		
Site mean palaeointensity 20.0 ± 3.2								Site mean VDM 3.5 ± 0.6			

Note: n, T_{int}: The number of points and the temperatures defining the linear segment; f: the percentage of the NRM used for the determination; g: an index of how well the NRM–TRM points are distributed along a straight line segment; q: quality factor defined by function $|b|^* f^* g/\sigma(b)$ where b is the slope and $\sigma(b)$ is its standard deviation (Coe et al., 1978); $F \pm \sigma$: palaeointensity estimate for individual samples and standard error; $M_{rs,480°C}/M_{rs,25°C}$: ratio of saturation remanence after heating to 480 °C to that of room temperature; $H_{c,480°C}/H_{c,25°C}$: ratio of coercivity after heating to 480 °C to room temperature; Fa ± σ_F: mean palaeointensity and its standard error for each lava; VDM ± σ_v: virtual dipole moment and standard error.

Thomas, 2003; Coe et al., 2004) or by thermal alteration during heating (Haag et al., 1995; Calvo et al., 2002). In order to determine the exact mechanism, more systematic rock magnetism measurements were carried out for samples with positive pTRM checks. First, twin samples were reheated in the palaeointensity oven using similar heating steps. Then, hysteresis parameters were measured at the room temperature for the thermal products (Henry et al., 2005).

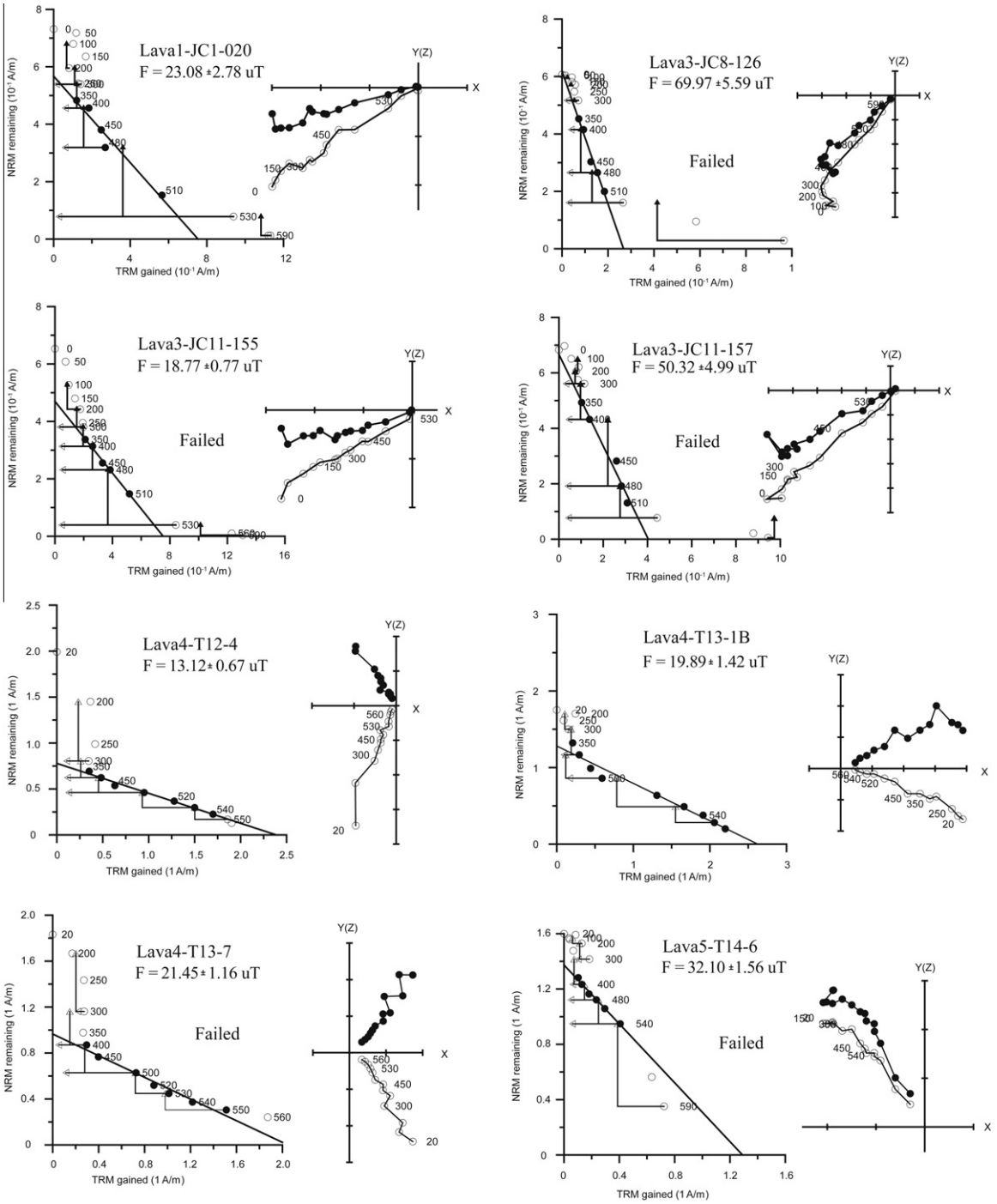


Fig. 7. Examples of significant linear NRM–TRM plots and orthogonal projections of the associated NRM demagnetization in sample coordinates. The TRM (pTRM-tail) checks are plotted on the NRM (TRM) axis with a triangle. Filled (open) circles denote data points accepted (rejected) from the best-fit line used to define the palaeointensity value. "Failed" indicates the palaeointensity determinations rejected by the new criterion of alteration.

The M_{rs} and M_s versus the treated temperatures are shown in Fig. 8. The curves can also be divided into three categories. The first category shows that M_{rs} and M_s gradually decrease with increasing temperature (Fig. 8a and b). The second category of samples shows that M_{rs} remains stable at temperatures <510 °C, and then increases sharply at elevated temperatures (Fig. 8c). In contrast, M_s changes less significantly than M_{rs} (Fig. 8d). The third category of samples show more complicated behavior. M_{rs} gradually decreases with increasing temperature up to 400 °C; it achieves minimum value at 480 °C and abruptly increases at higher temperatures (Fig. 8e). Meanwhile, M_s shows patterns comparable to that of the second category (Fig. 8f). This indicates that M_{rs} is more sensitive than M_s for indicating the possible thermal alterations in remanence.

To check the possible effects of thermal alterations during heatings on palaeointenstiy results, the ratio $M_{rs,480°C}/M_{rs,25°C}$ was proposed, where $M_{rs,25°C}$ and $M_{rs,480°C}$ are the room temperature saturation isothermal remanent magnetizations for the raw sample and the product one after a 480 °C thermal treatment, respectively. This ratio can be treated as an alteration index. When it approaches the unit, samples suffer insignificant thermal alterations.

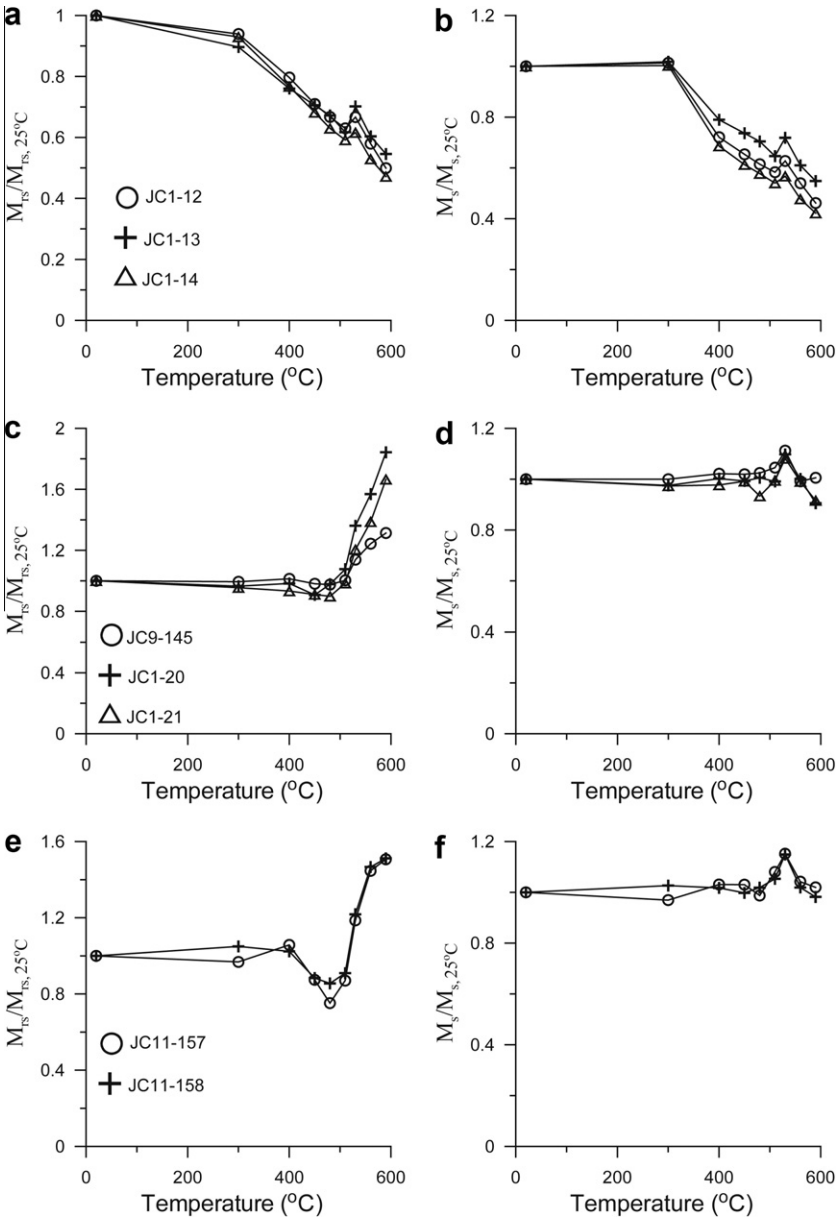


Fig. 8. The normalized room temperature saturation remanent magnetization (a, c, e) and the corresponding induce saturation magnetization (d, e, f) after heating to different temperature in palaeointensity furnace. $M_{rs,25°C}$ ($M_{s,25°C}$) is M_{rs} (M_s) for raw samples.

The correlation between the alteration index and the palaeointensity results is shown in Fig. 9. Generally, the data from Lava 1 and 2 show a negative correlation with some exceptions clustered on the left corner. Most of the data from Lava 5 are located in the left-lower conner. This strongly indicates that thermal alteration on the palaeointensity results is significant. To minimize such an effect, the alteration index between 0.9 and 1.1 was chosen as a new criterion, and 17 samples in three lavas meet this demand (Table 3). The average strength values per flow range from 16.4 ± 3.0 to 22.1 ± 2.4 μT, and with the corresponding VDMs are from $(2.7 \pm 0.5) \times 10^{22}$ to $(4.0 \pm 0.2) \times 10^{22}$ Am^2. The mean value of three lava flows is 20.0 ± 3.2 μT with corresponding VDM is $(3.5 \pm 0.6) \times 10^{22}$ Am^2.

8. Discussion

Although a stable ChRM was obtained from the majority of sample, it is crucial to consider the origin of magnetization. As discussed above, thermomagnetic analysis shows that the remanence is mainly carried by Ti-poor titanomagnetite or pure magnetite, which is probably resulted from high temperature oxyexsolution during the flow emplacement and indicates primary magnetization of thermoremanent origin (Buddington and Lindsley, 1964; Dunlop and Özdemir, 1997; Goguitchaichvili et al., 2002b). Thus, even though no fold and reversal tests are available, we conclude that the characteristic directions are probably primary. The ^{40}Ar/^{39}Ar dating of three lavas indicates the eruption of the studied lava sequences at Jianchang section occurred around ~119 Ma. The positive palaeodirection and ^{40}Ar/^{39}Ar dating confidently indicate that the lava flow erupted just at the onset of the CNS.

Arai diagrams with linear NRM–TRM curves are frequently encountered in palaeointensity studies, and pTRM check is a traditional approach to judge the chemical alterations during heatings in palaeointensity experiments. However, pTRM at T1 can only detect the magnetic particles with an unblocking temperature (T_{ub}) < T1 (Coe et al., 1984; Valet, 2003). If the newly-formed

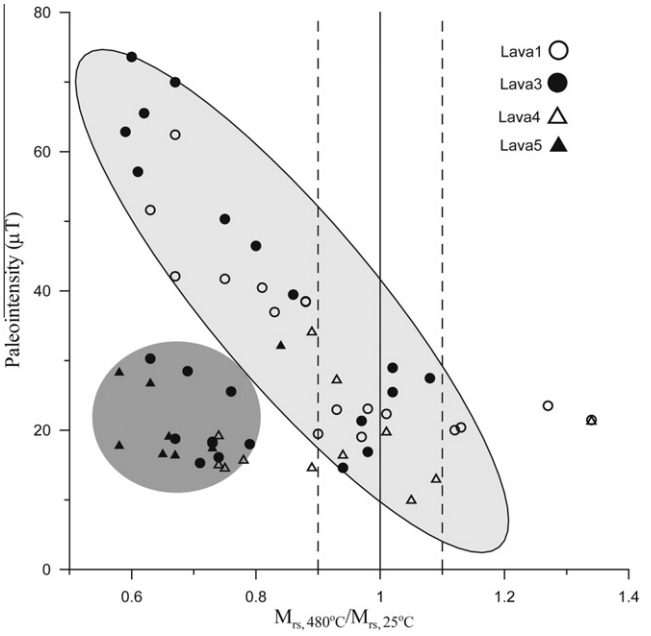


Fig. 9. Plot of palaeointensity results versus the ratio of saturation remanence after heating to 480 °C ($M_{rs,480°C}$) to that of room temperature ($M_{rs,25°C}$). The shaded areas classify the samples into two groups. The vertical line and two dashed lines indicate the region without significant thermal alterations.

magnetic particles have $T_{ub} > T1$, pTRM should still be positive although significant thermal alteration occurs. For such a case, the newly formed magnetic particles would increase the pTRM acquired at elevated temperatures, which results in a concaved-up pattern, and thus underestimate the palaeointensities. In contrast, when the primary magnetic minerals are transformed into a less magnetic state upon heating (e.g., from magnetite to maghemite, or maghemite to haematite), the corresponding pTRM will decrease, which will overestimate the palaeointensities. Such an alteration process can be detected by the parameter measured at the room temperature because it detects all magnetic minerals in the bulk sample after each treated temperature. Some studies on recent lava flows indicated that the Coe–Thellier method with pTRM check tends to yield higher values than the observed values (Hill and Shaw, 2000; Yamamoto et al., 2003; Mochizuki et al., 2004; Yamamoto and Hoshi, 2008). It is likely to be caused by the thermal chemical remanent magnetization (TCRM) acquisition during the initial cooling in nature (Yamamoto et al., 2003). And it is well known that pTRM tail checks are dependent on the angle between the NRM and the lab field. The pTRM tail check method is most effective when the applied field is perpendicular to the NRM (Yu and Dunlop, 2003). The direction of NRM demagnetization was affected by thermal alteration during heating can be used to judge the CRM effect in palaeointensity experiments (Mochizuki et al., 2004) if some magnetic phase with high blocking temperature was produced. However, it appears that this did not have significant effects on our samples except from Lava 5. The loss of NRM during heating may be a reduction of pTRM in NRM component, not due to demagnetization (e.g., Kosterov and Prévot, 1998; Mochizuki et al., 2004). It yielded a steep slope in the Arai plot, which were not detected by pTRM check and pTRM-tail check.

A negative correlation between the alteration index and the palaeointensity is suggested. However, if the magnetic domain state also changes accordingly, the pattern could be more complicated. This may partially explain some exceptional points in the lower left side in the Fig. 9. Regardless of the complexity, the results should be more reliable for samples with stable rock magnetic properties, especially at the temperature interval for the palaeointensity determinations (e.g., <510 °C in our study).

There has been a long-lasting debate on the Earth's palaeointensity during CNS (Fig. 10) (Sherwood et al., 1993; Juarez et al., 1998; Tarduno et al., 2001, 2002; Tanaka and Kono, 2002; Tauxe and Staudigel, 2004; Zhao et al., 2004; Zhu et al., 2004a,b; Shi et al., 2005; Granot et al., 2007; Tsunakawa et al., 2009). For example, low VDM(VADM) have been suggested from 3.6×10^{22} to 5.8×10^{22} Am^2 (Zhu et al., 2001; Pick and Tauxe, 1993; Riisager et al., 2003). It appears that there is an slight increasing trend from $\sim 3 \times 10^{22}$ to $\sim 5 \times 10^{22}$ Am^2 at the beginning of CNS. In contrast, fluctuated values from 4.0×10^{22} to 10.5×10^{22} Am^2 were also

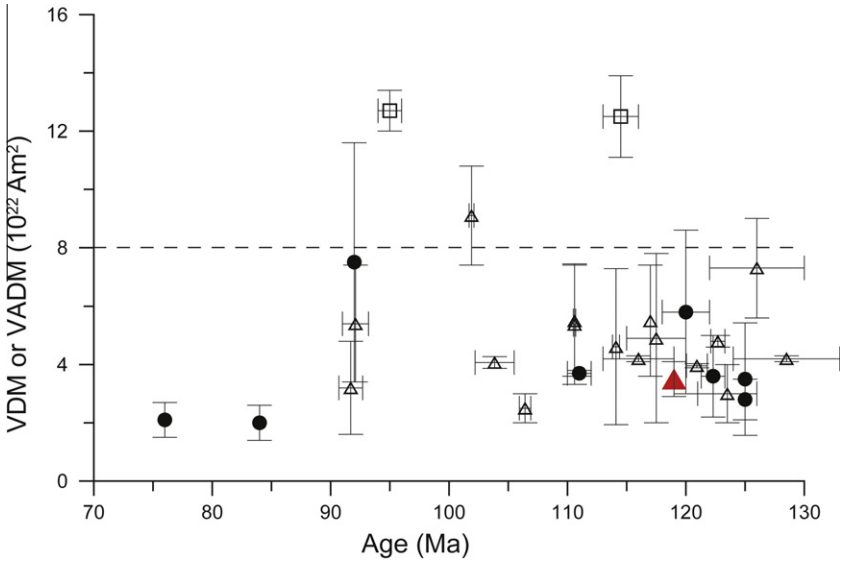


Fig. 10. Virtual dipole moment (VDM) or virtual axial dipole moment (VADM) within the Cretaceous. The horizontal and vertical bars are the errors of time and intensity of field, respectively, the red triangle is the palaeointensity of this study. Open square, single plagioclase crystal samples (Tarduno et al., 2001, 2002); solid circle, submarine basalt glass samples (Pick and Tauxe, 1993; Juarez et al., 1998; Riisager et al., 2003; Tauxe and Staudigel, 2004); open triangle, whole rock samples of the Cretaceous volcanics (Sherwood et al., 1993; Zhu et al., 2001, 2003, 2004a,b, 2008; Tanaka and Kono, 2002; Pan et al., 2004; Zhao et al., 2004; Shi et al., 2005; Ruiz et al., 2006; Granot et al., 2007; Hill et al., 2008); dashed line, the intensity of present-day field. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

reported (Goguitchaichvili et al., 2002a; Ruiz et al., 2006). Ruiz et al. (2006) further supported the hypothesis about the unreliability of 'Mesozoic Dipole Low' on the basis of the fluctuated palaeointensity pattern. However, in Ruiz et al. (2006)'s study, they also observed 'bizarre behavior' that the concave-up Arai-plot behavior with positive pTRM checks, but no explanation was given.

Clearly, such a pattern is consistent with our current study and can be reasonably explained by our new view that pTRM check at temperature T1 is not sensitive to the newly-formed magnetic minerals with T_{ub} > T1. If this is true, the fluctuated palaeointensities obtained by Ruiz et al. (2006) could also be affected by the thermal alteration as suggested in this study. Therefore, we strongly suggest that our new criterion could greatly improve the reliability of the palaeointenisity even with positive pTRM checks. By applying our new criterion, the high palaeointensities were excluded and the average of the palaeointensities associated with the most thermally stable samples is 3.5×10^{22} Am^2, which is consistent well with other four contemporaneous whole rock or SBG data (Pick and Tauxe, 1993; Zhu et al., 2001, 2003). Nevertheless, future studies on palaeointensity data are essential to further consolidate the onset of CNS.

9. Conclusion

The ideal magnetic carriers for palaeointensity studies are SD particles that can remain stable during heating in palaeointensity. However, the chemical alteration is common during heatings. Although pTRM check as a conventional method, is widely used to judge the chemical alteration, our results supports the idea that pTRM checks at certain temperatures (e.g., JC1) cannot detect the newly-formed magnetic particles with T_{ub} > T1. The newly proposed rock magnetic experiment can be an auxiliary criterion to further improve the reliability of the results. With the new criterion, our study shows that the palaeointensities just at the onset of CNS is lower than the present values. This improves our understanding of how the Earth's magnetic field evolved into the special state during CNS.

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